Universiti Kebangsaan Malaysia — The National University of Malaysia

UNIVERSITI WATAN KITA

FACULTY OF SCIENCE & TECHNOLOGY

KELAB STATISTIK UKM

DATA CHALLENGE — UKM STATISTICS CLUB

3 GOOD HEALTH AND WELL-BEING

4 QUALITY EDUCATION

8 DECENT WORK AND ECONOMIC GROWTH

Prepared by:
Ivy Witch

NAVIGATING THE INTERSECTION OF CAREER UNCERTAINTY & GEN-Z WELLBEING

Dayana | Afiqah | Najihah | Syasya | Aisha | Supervisor: Asst. Prof. Dr. Hafizah Bahaludin

PROBLEM STATEMENT

5.98M
Graduates in 2024

Despite record-high graduate output and **60% Labor Participation Rate**, the **market struggles persist.**

35.5%
Graduates remain in **low-skilled jobs**

Graduate Employment Challenges

Stagnant salaries, job insecurity, & mental health struggles affect many graduates

DASHBOARD

OBJECTIVES

- **To evaluate** the trend and pattern of graduate unemployment and underemployment in Malaysia from 2020-2024.
- **To identify** the statistical relationship between graduate employment and youth mental health by using Linear Regression between skill-related underemployment rates and DASS-21 distress scores.
- **To examine** parallel trends in graduate underemployment and youth mental health indicators.
- **To identify** potential mismatch hotspots by investigating which specific groups based on ethnicity, skill level or fields of study are being hit hardest by the utilization gap.

METHODOLOGY

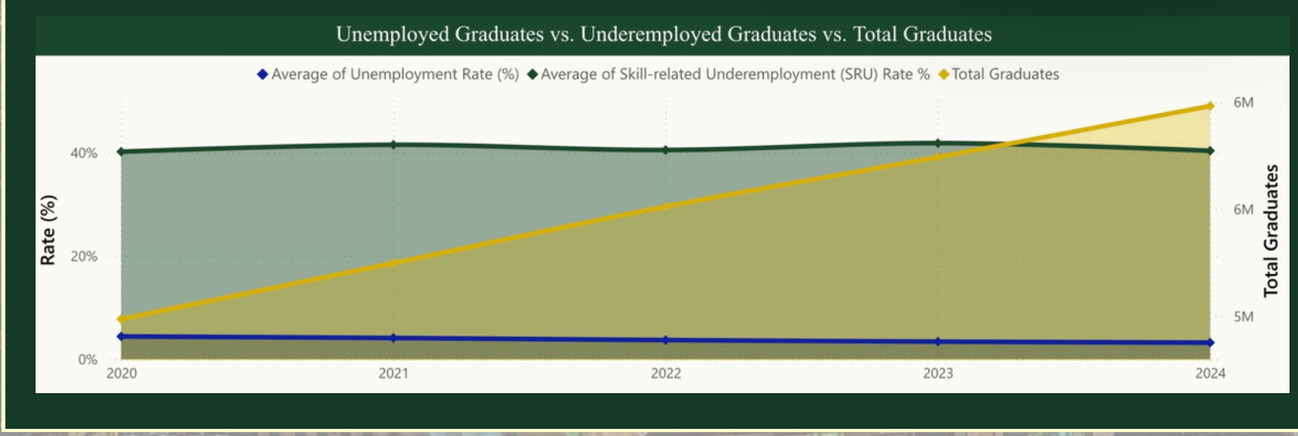
PHASE I — Data Synthesis & Preprocessing
- Data Collection (DOSM, MOHE, NHMS)
- Data Cleaning & Integration
- Metrics: Labor, Education & Health

PHASE II — Exploratory Data Analysis (EDA)
- Trend Analysis 2020-2024
- Utilization Gap Study
- Identify Patterns & Mismatches

PHASE III — Statistical Modelling
- Linear Regression
- Underemployment -> Mental Health
- Demographic Segmentation

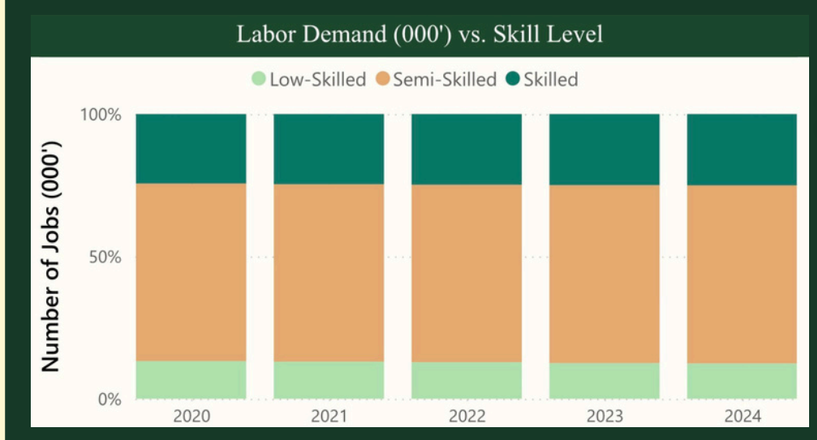
PHASE IV — Dashboard Visualization
- Power BI Dashboard
- Labor Mismatch & Mental Health Trends
- Policy Insights

FINDINGS



Vertical Mismatch

- Graduate output has been increasing over the years, reaching 5.98M in 2024
- Open Unemployment has trended toward stabilization and slight decline over the years, providing false sense of success
- The SRU rate remained high over the years, indicating that the labor market is absorbing graduates into the wrong tiers of employment



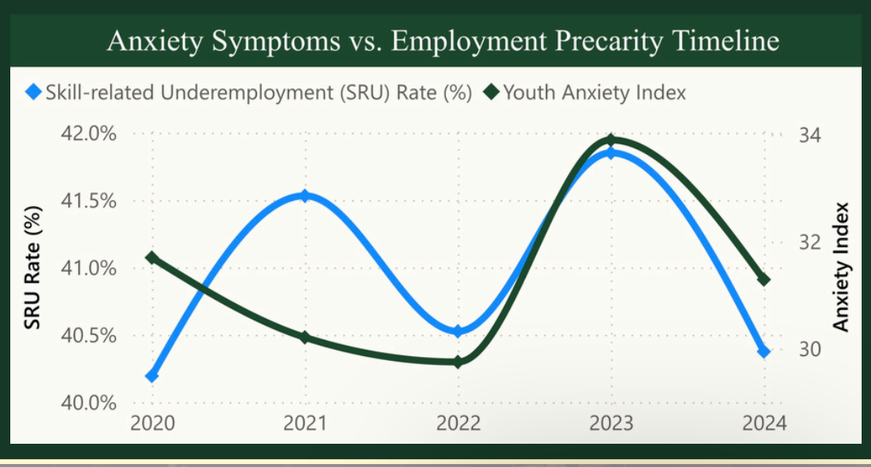
35.5% graduates work in semi-skilled or low-skilled roles

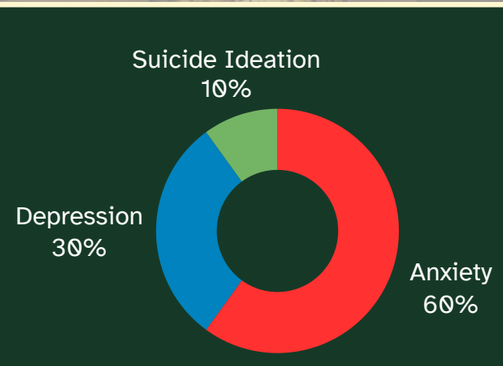
Critical Gap: 63.2% of unemployed graduates are Diploma holders.

Highest Risk Field: Agriculture & Veterinary Studies.

CATEGORY	HIGH-RISK SEGMENT	CRITICAL INSIGHT
Ethnicity	Others (Bajau, Iban, Bidayuh, Orang Asli, etc.)	Unemployment rate of almost half of the graduates population
Field of Study	Agriculture & Veterinary	The highest average unemployment rate
Graduate Proliferation Index (GPI)	STEM	GPI is lower than 1.00 (national average)
Level of Study	Diploma Holder	63.20% of unemployed graduates
Skill Level	Low-Skilled	Low demand and competitive salary

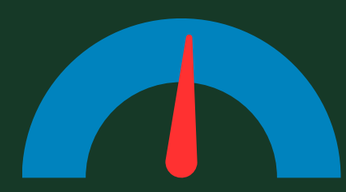
Peak Underemployment ≈ Peak Anxiety.





60% of graduates suffer from high anxiety symptoms

Stress Variance & Job Insecurity



>50% of Gen Z Graduates Stress Variance comes from Job Insecurity

CONCLUSION

- Job Gap: Unemployment is down, but job mismatch remains high.
- Stress Link: Working the "wrong" job directly triggers higher anxiety, stress, and depression.
- Mental Health Crisis: As underemployment rises, graduate wellbeing drops immediately.

Malaysia should prioritize high-skilled jobs and treat Youth Anxiety as a key measure of economic success.

"Redefining the graduate journey from surviving uncertainty to thriving in stability"